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EMERGING TECHNOLOGIES IN CONFLICT: THE IMPACT OF STARLINK IN THE RUSSIA – UKRAINE WAR

Laviniu BOJOR
"Nicolae Bălcescu" Land Forces Academy, Sibiu, Romania
laviniu.bojor@gmail.com

Tudorică PETRACHE
"Nicolae Bălcescu" Land Forces Academy, Sibiu, Romania
paratudor@yahoo.com

Cristian CRISTESCU
The 2nd Mountain Brigade "Sarmizegetusa", Brașov, Romania
ccristi2577@yahoo.com

ABSTRACT

This paper examines the role of technology on military communications, focusing on the impact of the commercial Starlink system on the battlefield. Utilizing the conflict in Ukraine as a case study, it will evaluate in detail how Starlink has influenced military actions, decision support, and the coordination of military efforts, offering a unique perspective on the transformation of communications in the context of modern conflicts. Furthermore, the study will analyze the essential role played by Starlink in supporting military morale and social resilience, highlighting how access to secure and rapid satellite communications has contributed to maintaining a positive mindset and cohesion under wartime conditions. By addressing these dimensions, the paper aims to capture the critical importance of emerging technologies not only in optimizing military actions but also in strengthening civil solidarity and social cohesion in times of crisis or war.

KEYWORDS: military communications, SATCOM, Starlink, social resilience, Ukraine war

1. Introduction

Communications have always played a crucial role in regional conflicts or major wars, evolving dramatically throughout history to meet the command-and-control needs of military leaders. Communication methods, such as the use of smoke signals and colored flags, now obsolete, were vital tools in the past for warning of imminent dangers or for coordinating military forces in the field. Messengers like the ancient runner Pheidippides (Lucas, 1976), covered great distances to mobilize Greek forces in the face of the massive Persian invasion or to announce the victory at Marathon in 490 BC (StMU Research Scholars, 2017). The lives of 200 American soldiers surrounded in World War I were saved through the ingenious use of the homing pigeon Cher Ami (George, 2023), which was the only option for Americans to deliver a message under conditions where their communication lines

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were cut and sending a courier was too risky. In World War II, communication technology made a significant leap with the introduction of the radio, which allowed for the efficient coordination of the tank-plane duo, marking the beginning of a new era in military operations leadership.

The radio was the technology that enabled Nazi Germany to implement the Blitzkrieg of World War II (Citino, 2008). Local tank-plane coordination, efficient regional communication between troops and command, or the remote direction of submarines in the Atlantic attacks are clear examples of the advantages that communication technology offers field military leaders. The evolution of military communications continues today through the introduction of advanced digital systems that allow secure and rapid communications, enabling long-distance data packet traffic such as text, voice, and video. Emerging technologies such as satellite communications, global positioning systems (GPS), and advanced data networks have enabled the implementation of concepts like "network-centric warfare", where decisions are made based on a continuously updated complete overview. The positive outcome of conflicts can no longer be achieved through the singular effort of the military domain, and survival on the battlefield must include both the military's capacity to endure on the front and the resilience of the population, as well as the engagement of diplomatic, economic, and social efforts.

2. Methodology

To deeply understand the impact of the Starlink system on military communications, we chose a scientific research methodology that combines qualitative data analysis with a case study on the use of Starlink in military actions in Ukraine starting from 2022. This approach will allow for understanding the role and evaluating the real impact that this technology has had on the ground, from both tactical and strategic perspectives. The qualitative analysis involved the collection and detailed examination of information from open sources (OSINT) and social media (SOCMINT), including scientific articles, reports, analyses, and specialized studies, as well as content uploaded to social media platforms such as YouTube, Telegram, or X (formerly known as Twitter). The paper aims to describe the Starlink technology and its technical features, as well as to clearly analyze how commercial emerging technologies have been used in military operations in Ukraine. This combination will contribute to a comprehensive understanding of the subject, highlighting not only the technological aspects but also the practical implications of using Starlink in the war zone.

3. Hybrid Warfare in Ukraine: Strategy, Impact, and the Role of Technology

The invasion of the Russian Federation that began on February 24, referred to by Russian propaganda as a "special military operation", has not achieved its objective of replacing the pro-Western government led by President Zelensky with a Kremlin-obedient government, and the "short" military intervention has transformed into a war of attrition that has now exceeded 2 years in duration. Russia's inability to occupy Kyiv led to a shift in Russian strategy, with troops being withdrawn from the northern part of Ukraine to focus military efforts in the South and Southeast regions. Simultaneously, it initiated widespread attacks using its military arsenal (ground-to-ground missiles, artillery, kamikaze drones, etc.) on Ukraine's energy system. This effort could not be considered accidental, given the statement from October 2022 by Ukraine's energy minister, who mentioned that approximately 30% of Ukraine's energy infrastructure was attacked in a single day. In the seven months of war, there was no energy facility that was not hit by Russian missiles or drones (Talya, n.d.), a vulnerability occurred by the fact that both countries were part of the Union of Soviet Socialist

Republics (USSR) until 1991, and infrastructure maps allow for the identification of vulnerable points in the energy networks (Birnbaum et al., 2022).

In September 2022, before the arrival of winter, Russia launched a wave of aerial attacks against Ukraine's energy infrastructure. A single attack involved 43 cruise missiles (Dickinson, 2023), and this process of targeting Ukrainian energy facilities continued throughout 2023. The attack on the electricity grid represents an active component of the hybrid conflict, an extremely complex concept with implications and conditions in multiple domains (Tudorache et al., 2023), including energy and communications. The attacks on these critical infrastructures directly disrupt the activities of the population, economic agents, non-governmental organizations, and state institutions (Cîrdei, 2018). Russian attacks have destroyed about 61% of Ukraine's electricity generation capacity and damaged over 50% of the country's energy sector in general. Affected areas had access to electricity, heating, and internet for only a few hours a day, and hospitals, businesses, and households were forced to resort to diesel generators for electricity production.

A 2022 UN report highlighted that the war in Ukraine has caused infrastructure damages exceeding 100 billion dollars, with the risk of 90% of the country entering extreme poverty and vulnerability (Ukraine War, 2022). Furthermore, attacks on Ukraine's energy infrastructure have caused over 10 billion dollars in damages, leaving over 12 million people without access or with limited access to electricity and disrupting water supply and heating systems. Not only the energy system was attacked. Water networks, public administration buildings, bridges, railway stations, hospitals, schools, theaters, and even residential buildings were attacked by missiles and Iranian drones from the first year of conflict. Russia's invasion of Ukraine has caused extensive destruction, with damages that cannot be evaluated at this moment. Estimates for the reconstruction of Ukraine at the end of the first year of terror were approximately 350 billion dollars (Ovaska, 2022). These have estimated costs for land demining, repairing schools, apartments, and affected hospitals. But they cannot be realistic since the conflict has just entered its third year and shows no signs of ending. This strategy, known as "Strategic bombing", has been used since World War II in campaigns conducted in Nazi Germany, Japan, North Korea, Vietnam, or Yugoslavia. It aims to reduce military and industrial capacities to produce and supply troops on the front with weapons and equipment and serves as an instrument of pressure on the population with the goal of destroying morale and the desire to resist.

Strategic bombing campaigns have also targeted communication systems, both military and civilian, and mobile communication towers were no exception. In the case of the energy grid and other industrial infrastructures located far from the front line, repair works, and the national effort adapted to the wartime context have managed their partial or total restoration, to provide the population with minimal social services. However, on the front line, they are within the range of Russian artillery, making the repair and restoration of mobile communication systems dangerous and the mobile phone network unable to be reinstated. Combined attacks, with missiles/drones and cyber-attacks, led to the shutdown of the internet within up to 25 km from the front line (Palli, 2022). The destruction of the communication system aimed to isolate troops from higher military echelons and cut off connections between soldiers and their families. Ukrainian efforts to survive the massive Russian attack included requests for international support, both military and civilian. A tweet on the social network X posted by the Minister of Digital

Transformation Mykhailo Fedorov (Mykhailo Fedorov [@FedorovMykhailo], 2022) received an unexpected response from Elon Musk, the founder of SpaceX. The CEO of the company agreed online to support Ukraine in its effort to resist, and 2 days later, Starlink's satellite internet services were active, and the equipment received provided data to both soldiers and the population affected by the war.

4. Starlink: Deploying Satellite Connectivity in War Zones

Starlink is a satellite constellation operated by the American civilian company SpaceX, which, according to its official website, offers coverage in over 70 countries (Starlink, 2024). The main purpose of Starlink is to provide satellite internet access in areas where traditional infrastructure and connectivity are limited or non-existent. SpaceX began launching Starlink satellites in 2019, and these satellites orbit in low Earth orbit at an altitude of approximately 550 km above the Earth. The latency (delay in communication time) for Starlink varies between 25 and 60 ms on land and 100+ ms in remote locations (such as oceans, islands, Antarctica, Alaska, northern Canada, etc.). Download speeds for Starlink users are between 25 and 220 Mbps, with most experiencing speeds over 100 Mbps, and upload speeds are between 5 and 20 Mbps.

Comparisons made by communication analysts regarding the differences between satellite and terrestrial communications identify key aspects between the two systems (Table no. 1).

Table no. 1

Comparison between Satellite and Terrestrial Communications

Parameters	Satellite Communication	Terrestrial Communication
Communication medium	Satellites installed in space	Physical infrastructure on the ground (Cables, fiber optic, microwave links)
Coverage	Global coverage wide across $(1/4)^{th}$ of the earth	Limited to specific geographical regions or within range of physical infrastructure
Latency	Higher as signals travel longer distances (120 ms from earth to satellite and 120ms from satellite to earth back)	Lower as signals travel shorter distances
Susceptibility to Signal interference	Prone to atmospheric conditions and signal blockage	Vulnerable to physical disruptions and electromagnetic interference
Mobility	Suitable for remote and mobile communication	Mostly fixed and limited mobility
Initial cost	Higher	Lower

Parameters	Satellite Communication	Terrestrial Communication
Data speed	Usually lower compared to some terrestrial options	Higher especially using advanced fiber optic networks
Reliability	Susceptible to space related issues such as solar radiation	More reliable in stable environments with periodic maintenance
Examples	GPS, satellite TV, Satellite internet, Satellite telephony	Cellular networks, Wi-Fi, Ethernet, fiber optic cables

(Source: RF Wireless World, n.d.)

Terrestrial communications have high transfer speeds, lower latencies, are less vulnerable to atmospheric conditions or to signal jams, have reduced costs, and are more efficient in terms of maintaining equipment and network connections. However, in the context of the conflict in Ukraine, where the terrestrial architecture is destroyed or permanently attacked, the satellite alternative becomes not a better option but the only one. The inclusion of Tesla Powerwall batteries alongside the mobile Starlink equipment solves both the mobility issue and the lack of the electric grid in contact areas (Opinion | 'How Am I in This War?, 2023). Using a constellation of thousands of satellites, Starlink has provided Ukraine with the necessary equipment to connect to the internet and escape the "black hole" that the Russian Federation created by destroying terrestrial communications.

5. Starlink Role in Ukraine War

From the first days of activation, SpaceX's service has been utilized by both civilians and military personnel. By May 2022, over 150,000 Ukrainians were using Starlink daily, and Minister Fedorov publicly stated that Starlink "*is crucial support for Ukraine's infrastructure and the restoration of destroyed territories*" (Jayanti, 2023). We aim to identify the main actions involving the use of Starlink technology in Ukraine:

– Command and Control, Intelligence, Surveillance, and Reconnaissance (C2ISR): Starlink's satellite communications enabled Ukrainians to command and control their units on the battlefield through efficient coordination of forces and resources. Starlink allowed Ukrainian military forces to restore and maintain military communications in areas where terrestrial infrastructure was destroyed or inaccessible. By providing access to high-speed internet, Starlink facilitated the rapid collection and distribution of data and information, enhancing the Intelligence Preparation of the Battlefield (IPB). Satellite communications provided by Starlink maintained vital links with forces under siege in the Azovstal steel plant's underground shelters (Schwirtz, 2022). Trapped in Mariupol, the besieged Ukrainian troops were able to maintain contact with the outside world with the help of hard-to-jam Starlink equipment. Images and videos transmitted, or even live interviews with the wounded captives in the Azovstal plant's undergrounds, went global. From the invasion's first days, this contributed to the perception that Ukrainian forces were not yielding and could resist the much stronger Russian military machine, boosting the morale of the troops and the civilian population.

– Support for the Population: Beyond military advantages, Starlink can provide vital communications for the population caught in conflict. The lack of communication affected emergency services. Satellite communications offered humanitarian support to the civilian population in both conflict-affected and rural areas lacking terrestrial communications. The internet allowed the population to receive real-time warnings about missile attacks or the availability of shelters in the area. The connectivity provided by Starlink turns this commercial technology into a strategic national tool due to its proven utility during the government's effort to manage both the Russian invasion and the vital services for the civilian population (Dalmia & Mittal, 2024).

– Drone Attacks: Starlink terminals enable the operation of unmanned aerial and maritime vehicles in reconnaissance or remote attack missions, ensuring real-time control and data transmission without significant delays. Bayraktar UAVs, notable for their unmanned operation, played a crucial role in the conflict, evidenced by a song dedicated to these UAVs by Ukrainian soldiers (YouTube – Kronika24.pl, 2022). Although their active deployment has diminished due to enhanced Russian anti-air defenses, their place has been taken by small, commercially available drones. These drones are hard to detect on radar but can carry out bombing devices or acting as guided kamikaze UAVs with the assistance of Starlink. Satellite communications have proven their value in the Black Sea area as well. Though left without a naval fleet, Ukraine managed to mobilize and develop maritime drones that damage or sink Russian ships in the Black Sea. Using Neptune and Harpoon anti-ship missiles, alongside kamikaze drones like "SEABABY" or "MAGURA" (Sutton, 2022), they took out the Russian naval pride Moskva (Romaniuk, 2022), and ships like Ivanovets (The Times and The Sunday Times, 2024), Olenegorsky Gornyak (Inocencio, Smith & Falk, 2023), Vasily Bykov, or Sergey Kotov (Suchomimus, 2023). Ukraine has managed, with the help of drones coordinated through Starlink, to limit or even eliminate the Russian Federation's freedom of action in the Black Sea area. Furthermore, satellite communications not only offered pilots the chance to guide drones to the target but also filmed the actions and used them in propaganda efforts meant to boost the morale of the troops by portraying the Russian military as weak and easily countered.

– Social Resilience: While most analyses prioritize the kinetic military actions on the ground, like target acquisition or the coordination of troops and unmanned systems, Starlink communications have provided Ukraine with the necessary ingredient to withstand the powerful Russian army: soldiers' connection with their families. Social resilience is defined as the ability of people and communities to cope with external stress and shocks, to withstand disasters, and to recover from post-disaster (Kwok et al., 2016). Surveys and interviews with respondents from Ukraine (regions of Kyiv, Lviv, Zakarpattia, Mykolaiv, Sumy, Chernihiv, and Dnipropetrovsk) confirm the crucial role of family as a factor of social resilience in the face of war challenges. The results indicate that 59% of respondents believe the situation in Ukraine will improve, and 67.2% consider this due to family support (Kostenko et al., 2024). The dual soldier-family communication was exclusively due to Starlink. The lack of the energy grid and the destruction of mobile communication towers isolated front-line soldiers, and the absence of information about their families would have lowered troop morale at the contact line. Similarly, the support of local or international communities would not have existed without the small victories transmitted from the field. Satellite technology allowed the Ukrainian civil society to emerge from the initial shock that could have paralyzed the country in a manner similar to the occupation of the Crimean Peninsula in 2014. The Russian military's failures were widely distributed on social media, and internet access created a general sentiment that Ukraine could resist. Videos of Ukrainian troops' victories were used by Zelensky in strategic communication, insisting on the idea that both the president and Ukraine stand and fight, not yielding. Starlink enabled the distribution of media content created from the trenches, showing soldiers dancing (The Independent, 2022) despite the mud or snow, soldiers smiling despite the constant artillery bombardment (The Telegraph, 2023), soldiers rescuing civilians or pets affected by the war (YouTube – AFP News Agency, 2022), or soldiers defying death in the face of an aggressor that does not respect human rights or international humanitarian law rules through recorded abuses such as killing civilians or executing a prisoner of war (YouTube – Sky News, 2023). At the same time, soldiers had access to the reaction of Ukrainian communities in occupied areas. They saw that their efforts were not isolated, as the population resisted illegal occupation and attempted to counteract the so-called 'demilitarization and denazification' of Ukraine. Civilians protested against Russian military vehicles, offered poisoned food to invaders, or flooded areas to prevent access to certain locations (Bojor & Cîrdei, 2023). The live transmission of the war in Ukraine on social media, facilitated by Starlink's satellite communications, successfully connected and mobilized both national civilian communities distant from the front line and international attention, compelling action against Russia's unprovoked aggression.

These insights underscore Starlink's essential role in the Ukraine conflict, highlighting the capacity of commercial technology to influence the dynamics of contemporary military conflicts.

6. Conclusions

Emerging disruptive technologies can have a significant impact in the context of modern conflicts, providing rapid and efficient solutions such as restoring and maintaining military communications. Through its support to Ukraine, Starlink technology not only facilitated military and civilian communications in the conflict zone but also provided the means to combat disinformation and isolation strategies imposed by the Russian Federation. It mobilized the masses and international support in an economic and military effort that enabled Ukraine to resist or even regain the initiative in certain conflict zones such as Kharkiv. The work validates the viability of integrating civilian technological innovations into the equation of contemporary conflicts, while also emphasizing the imperative need for a consolidated synergy between the defense sector and the private sector to ensure swift and effective intervention in crisis or wartime scenarios.

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